

FIELD THEORY PART-II

Phase 5: Magnetostatics Core

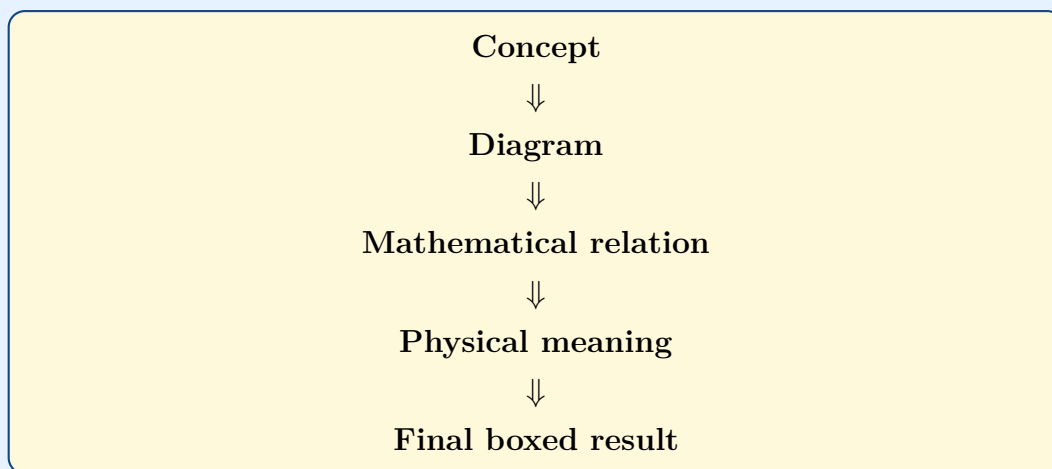
Biot–Savart law, magnetic vector potential and loop numericals

Academic Article Style Notes for Semester Preparation

How to Use This Phase

This phase is made for semester preparation from a very basic level. Read the concept first, then understand the diagram, then learn the derivation pattern, and finally practise the exam-style writing block.

The recommended writing order is:



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1 Aim of Phase 5

Main Goal

This phase covers the magnetostatics core: Biot–Savart law, magnetic vector potential, loop numericals, and magnetic energy density.

2 What Is Magnetostatics?

Meaning

Magnetostatics deals with magnetic fields produced by steady currents.

Main Assumption

In magnetostatics:

$$\frac{\partial}{\partial t} = 0.$$

This means field quantities do not change with time.

3 Biot–Savart Law

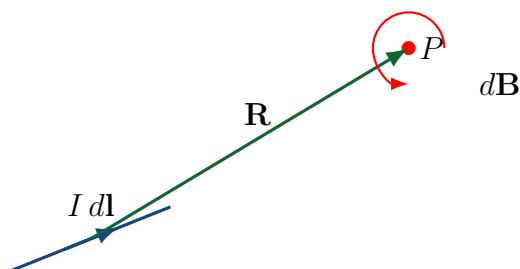
Statement

For a current element $I d\mathbf{l}$, the magnetic flux density at a point is:

$$d\mathbf{B} = \frac{\mu I}{4\pi} \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2}$$

or:

$$d\mathbf{B} = \frac{\mu I}{4\pi} \frac{d\mathbf{l} \times \mathbf{R}}{R^3}$$



Current element produces magnetic field at point P

Figure 1: Current element used in Biot–Savart law.

4 Magnetic Field Intensity

Relation Between B and H

$$\mathbf{B} = \mu \mathbf{H}$$

Therefore:

$$d\mathbf{H} = \frac{I}{4\pi} \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2}$$

5 Magnetic Vector Potential

Definition

Magnetic vector potential $\mathbf{A}$ is defined by:

$$\mathbf{B} = \nabla \times \mathbf{A}$$

Why It Is Possible

Since:

$$\nabla \cdot \mathbf{B} = 0$$

and:

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0,$$

we can express $\mathbf{B}$ as curl of another vector $\mathbf{A}$.

6 Rectangular Loop Formula

Axial Field of Rectangular Loop

For a rectangular loop with half-sides a and b , carrying current I , the magnetic flux density at axial distance h is:

$$B = \frac{\mu_0 I ab}{\pi \sqrt{a^2 + b^2 + h^2}} \left[\frac{1}{a^2 + h^2} + \frac{1}{b^2 + h^2} \right]$$

Magnetic Field Intensity

Since $B = \mu_0 H$ in air:

$$H = \frac{I ab}{\pi \sqrt{a^2 + b^2 + h^2}} \left[\frac{1}{a^2 + h^2} + \frac{1}{b^2 + h^2} \right]$$

7 Square Loop on Central Axis

Square Loop Formula

For square loop with half-side a :

$$B = \frac{2\mu_0 I a^2}{\pi(a^2 + h^2)\sqrt{2a^2 + h^2}}$$

8 Magnetic Energy Density

Formula

Magnetic energy density is:

$$w_m = \frac{B^2}{2\mu}$$

or:

$$w_m = \frac{1}{2}\mu H^2$$

9 Exam Numerical Pattern

How to Solve Loop Numerical

Write:

1. Given dimensions and current.
2. Convert all dimensions to metre.
3. Draw loop and point.
4. Identify half-sides.
5. Use correct formula.
6. Calculate B or H .
7. If asked, calculate energy density.

Common Mistake

Do not use full side length in formula when the formula uses half-side a and b .

10 Final Self-Test for This Phase

Instructions

Answer these without looking at the notes. If you cannot answer a point, revise that section again before moving to the next phase.

1. Can I state the main definition of this phase?
2. Can I draw the main diagram neatly without overlap?
3. Can I explain the physical meaning in two or three lines?
4. Can I write the final formula with correct symbols?
5. Can I connect this topic with at least one past-year question pattern?
6. Can I write a semester-style answer with definition, diagram, derivation and final result?

Completion Rule

This phase is complete only when the concept, diagram, formula and answer-writing pattern are all ready.